Candidate seat No: _______________

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Third Semester of B.Tech. Theory Examination(EC)

March-April 2018

MA243 Transforms and Matrix Algebra

Date: 26/03/2018 (Monday) **Time: 01:30 pm to 04:30 pm** **Maximum Marks: 70**

Instructions:

i. The question paper contains two sections.
ii. Section I and II must be attempted in separate answer sheets.
iii. Figures to the right indicate marks.
iv. Make suitable assumptions and draw neat figure where it is required.
v. All notations and terminologies are standard.

Section-I

Q-1 **Choose the correct answer from the given options in the following:** **[07]**

(1) The period of $\sin x$ is________. **01**

(a) π **(b)** $-\pi$ **(c)** 2π **(d)** -2π

(2) The value of an integral $\int_0^{2\pi} \cos x\, dx$ is_________. **01**

(a) 0 **(b)** 1 **(c)** -1 **(d)** 2π

(3) Which one of the following is an even function of t? **01**

(a) t^2-3t. **(b)** t^2. **(c)** $\sin(5t)$ **(d)** t^3-5.

(4) The value of $L\left[f(t-a)u(t-a)\right]$ is_______. **01**

(a) $e^{as}F(s)$ **(b)** $e^{-as}F(s)$ **(c)** $e^{at}F(t)$ **(d)** $e^{-at}F(t)$

(5) The value of $L\left[\sinh\left(2t\right)\right]$ is________. **01**

(a) $\frac{2}{s^2-4}$ **(b)** $\frac{2}{s^2+4}$ **(c)** $\frac{2}{s^2-2}$ **(d)** $\frac{2}{s^2+2}$

(6) The Laplace transform of a function f exists if the function **01**

(a) f is piecewise continuous.
(b) f is of exponential order.
(c) f is piecewise continuous and of exponential order.
(d) f is neither piecewise continuous nor of exponential order.

(7) The inverse Fourier transform of a function f is **01**

(a) $f(x)=\frac{1}{\sqrt{2\pi}}\int_0^{\infty} F(s)e^{isx}ds$.

(b) $f(x)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} F(s)e^{isx}ds$.

(c) $f(x)=\frac{1}{\sqrt{2\pi}}\int_0^{\infty} F(s)e^{-isx}ds$.

(d) $f(x)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} F(s)e^{-isx}ds$.

Q-2 Attempt any Four. **[16]**

(1) Find the Fourier series of $f(x)=\dfrac{\pi-x}{2}$ in the interval $[0,2\pi]$. **04**

(2) Expand $f(x)$ in a Fourier series, where $f(x)=\begin{cases}0 & ; \ -2<x<0 \\ 1 & ; \ \ 0<x<2\end{cases}$. **04**

(3) Obtain the Fourier series of $f(x)=x^2$, where $-1<x<1$. **04**

(4) Using Convolution theorem find the inverse Laplace transform of $\dfrac{1}{s^2(s-1)}$. **04**

(5) Find the Laplace transform of the function $f(t)$ defined by $f(t)=\begin{cases}3 & ; \ 0<x<2 \\ 0 & ; \ 2<x<4\end{cases}$, $f(t+4)=f(t)$. **04**

(6) Evaluate $\displaystyle\int_0^\infty \frac{x^2}{(x^2+1)^2}dx$ using Parseval's identity for the Fourier transform. **04**

Q-3 (a) Attempt any Two. **[06]**

(1) Define the half range Sine and half range Cosine Fourier series. **03**

(2) Solve the differential equation $\dfrac{d^2y}{dt^2}+y=1$ with $y(0)=1$ and $y'(0)=0$ using Laplace transform. **03**

(3) Find the Laplace transform of $t^3+3t^2+\sin(2t)+e^t\cos(2t)$. **03**

Q-3 (b) Define the Fourier transform and find it for the function **[06]**

$$f(x)=\begin{cases}1-x^2; & |x|\le 1 \\ 0 \quad ; & |x|>1\end{cases}.$$

OR

Q-3 (b) Define the Fourier sine and cosine transform. Find the Fourier sine and cosine transform of $f(t)=e^{-at}$. **[06]**

Section-II

Q-4 Choose the correct answer from the given options in the following: **[07]**

(1) If $\begin{bmatrix}1\\1\end{bmatrix}$ is an eigenvector of a matrix $\begin{bmatrix}1 & 4\\3 & 2\end{bmatrix}$, then the corresponding eigenvalue is_______. **01**

(a) 5 **(b)** 2 **(c)** -2 **(d)** Cannot be determined

(2) If 0 is an Eigen value of a matrix A, the value of $\det(A)$ is____________. **01**

(a) 1 **(b)** 0 **(c)** 2 **(d)** -1

(3) The Z-transform $U(z)$ of a discrete time signal $u(n)$ is defined as **01**

(a) $\sum_{n=-\infty}^{\infty} u(n)z^{n}$ **(b)** $\sum_{n=-\infty}^{\infty} u(n)z^{-n}$ **(c)** $\sum_{n=0}^{\infty} u(n)z^{n}$ **(d)** None of the mentioned

(4) If $U(z)$ is the Z-transform of signal $u(n)$, then the Z-transform of $\delta(n)$ is__________. **01**

(a) 1 **(b)** 0 **(c)** -1 **(d)** Not possible

(5) According to the shifting property of Z-transform if $U(z)$ is the Z-transform of signal $u(n)$, then what is the Z-transform of u_{n-k} ? **01**

(a) $z^{k}U(z)$ **(b)** $z^{-k}U(z)$ **(c)** $U(z-k)$ **(d)** $U(z+k)$

(6) If $r = x\hat{i} + y\hat{j} + z\hat{k}$, then div r equal to ______________. **01**

(a) 3 **(b)** r **(c)** 0 **(d)** -r

(7) If a vector $a = \frac{i}{\sqrt{3}} + \frac{j}{\sqrt{3}} + \frac{k}{\sqrt{3}}$, then modulus of vector a is ________________. **01**

(a) 1 **(b)** 0 **(c)** -1 **(d)** 2

Q-5 Attempt any Four. **[16]**

(1) Find the eigenvalues and eigenvectors for the matrix $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$. **04**

(2) Using Cayley -Hamilton theorem reduce $2A^5 - 3A^4 + A^2 - 4I$ into a linear polynomial in A, where $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$. **04**

(3) Find Z-transform of $\sin n\theta$ and $\cos n\theta$. **04**

(4) Find the Z-transform of $(i)\ a^n$ $(ii)\ (n+1)^2$. **04**

(5) Find the inverse Z-transform of $\frac{4z}{z-a}$ for (i) $|z| > |a|$ (ii) $|z| < |a|$. **04**

(6) If $\overline{r} = (a\cos\eta t)\hat{i} + (b\sin\eta t)\hat{j}$, where a, b and η are constants then prove that $\frac{d^2\overline{r}}{dt^2} + \eta^2\overline{r} = 0$. **04**

Q-6 **(a) Attempt any Two.** **[06]**

(1) Evaluate Curl of the vector $\overrightarrow{F} = xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k}$ at (1, -1, 1). **03**

(2) A particle moves along the curve $x = 2t^2$, $y = t^2 - 4t$, $z = 3t - 5$,where t is the time. Find the component of it's velocity and acceleration at time t=1. **03**

(3) Find the directional derivative of $f(x, y, z) = 2x^2 + 3y^2 + z^2$ at point (2,1,3) in the direction of the vector $a = i - 2k$. **03**

Q-6 **(b)** Find the characteristic equation of a matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence obtain $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$. **[06]**

OR

Q-6 **(b)** Find the eigenvalues and eigenvectors for the matrix $A = \begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$. **[06]**

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